FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 20-09-2020

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

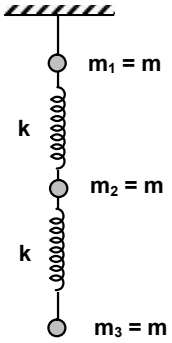
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Three identical balls of equal mass (m) are suspended vertically from a thread and two springs of same spring constant k as shown in the figure. The distance between the first and second ball and the second and third ball are same at equilibrium. The centre of mass of the whole system coincides with the second ball. The thread supporting the upper ball is cut and the system starts to fall freely. At some later time, it is observed that both the springs acquire their natural lengths. (Assume this happens) Find the distance of the centre of mass of the system from the second ball at that instant.



(A) $\left(\sqrt{3}-1\right)\frac{mg}{2k}$

(B) $\frac{mg}{3k}$

(C) $\frac{mg}{2k}$

(D) zero

2. Two electrons initially at rest and having a charge e and mass m each, are placed at a separation d apart. They are given identical speeds v in opposite direction in the same plane and it is observed that in the subsequent motion they move so that the distance between them remain constant at d. A uniform homogeneous magnetic field(B) is present perpendicular to the plane, Ignoring the magnetic effects of the moving charges on each other, identify the correct option for the separation between the charges to remain constant.

(A) The minimum distance between the electrons should be $d_{min} = \sqrt{\frac{1}{4\pi\varepsilon_0 evB}}$

(B) The minimum distance between the electrons should be $d_{min} = \left[\frac{m}{2\pi\varepsilon_0 B^2}\right]^{1/3}$

(C) The minimum distance between the electrons should be $d_{min} = \sqrt{\frac{1}{2\pi\varepsilon_0 evB}}$

(D) The minimum distance between the electrons should be $d_{min} = \left[\frac{2m}{\pi\varepsilon_0 B^2}\right]^{1/3}$

3. A diatomic molecule having atoms of masses m_1 and m_2 and its potential energy function about the equilibrium position r_0 is given by $U(r) = -A + B(r - r_0)^2$, where A and B are positive constants. When the atoms vibrate at high temperature condition, the square of angular frequency of vibration will be

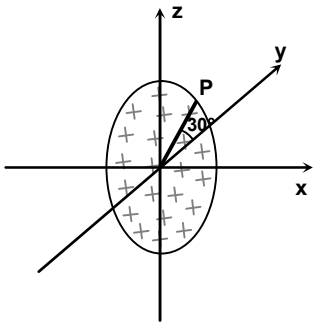
(A) $\frac{2B}{m_1}$

(B) $\frac{2B}{m_2}$

(C) $\frac{2B(m_1 + m_2)}{m_1 m_2}$

(D) $\frac{B(m_1 + m_2)}{2m_1 m_2}$

4. A uniformly charged non-conducting circular disc is placed in the y-z plane with its centre at origin. Its surface charge density is σ C/m^2 and radius R meter. A unit positive charge is brought very slowly from infinity to the point P on its circumference (as shown in the figure). The work done (in Joule) by the external agent in this process is



(A) $\frac{\sigma R}{2\varepsilon_0}$

(B) $\frac{\sqrt{3}\sigma R}{2\varepsilon_0}$

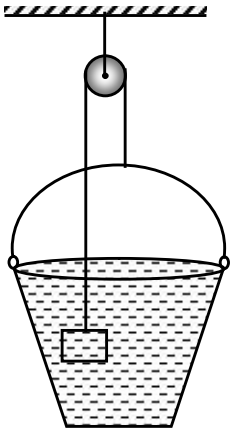
(C) $\frac{\sigma R}{\sqrt{3}\varepsilon_0}$

(D) $\frac{\sigma R}{\pi\varepsilon_0}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. A solid iron cube of volume 2×10^{-3} m^3 is fastened to one end of a light cord and the other end of the cord is attached to a light plastic bucket containing water. The cord has negligible mass and passed over the pulley and the iron cube is suspended in water as shown in the figure. It is found that the system is in equilibrium. If density of iron and water is 7.8 gm/cc and 1 gm/cc respectively, then



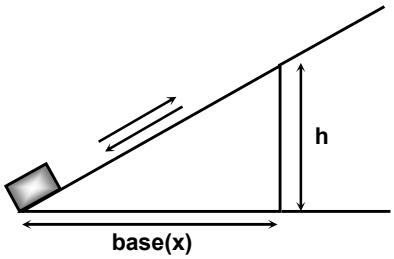
(A) volume of water in the bucket is 11.6 litre

(B) volume of water in the bucket is 13.6 litre

(C) If more water is very slowly poured in the bucket then the maximum additional volume of water that can be poured in the bucket without disturbing the equilibrium of the system is 4 litres.

(D) If the whole water in the bucket is allowed to evaporate then the final tension in the string is 7.8 kg-wt.

6. A block of mass 2 kg is placed on the inclined plane on which it does not spontaneously start to slide. It is pulled up and then down the incline to the initial position very slowly in such a manner that the applied force is always parallel to the slope. (Refer to the figure). The work done by the external agent is 40 Joule. Assume co-efficient of static and kinetic frictions are same.



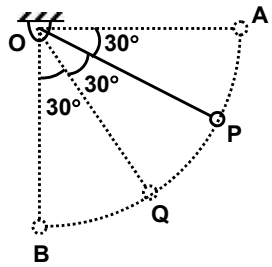
(A) The maximum possible value of h is 0.5 m

(B) The maximum possible value of h is 1 m

(C) For a given value of total work done by the external agent, the length of the incline's base depends on the co-efficient of friction.

(D) For a given value of total work done by the external agent, the length of the incline's base does not depend upon the angle of inclined plane.

7. A simple pendulum is released from rest from point A with its string horizontal. It moves along the path APB as shown in the figure. If t_1, t_2 and t_3 be the time taken to move from A to P, P to Q and Q to B respectively. Then,



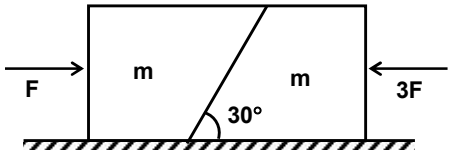
(A) $t_1 > t_2 + t_3$

(B) $t_1 < t_2 + t_3$

(C) $t_1 < t_2 < t_3$

(D) $t_1 > t_2 > t_3$

8. Two blocks each of mass m = 3kg move under the application of forces F and 3F as shown in the figure. The blocks do not slide with respect to each other and all of contact surfaces are smooth. The normal reaction between the blocks can have the value/values.



($g = 10\ m/s^2$)

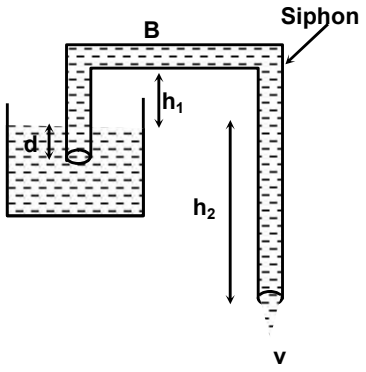
(A) 35 N

(B) 30 N

(C) 20 N

(D) all the above

9. A liquid of density ρ = 1gm/cc is taken out of the container (shown in the diagram) by a siphon. The liquid comes out with a speed v. h_1, h_2 and d are as shown in the diagram. Assume the liquid to be ideal and atmospheric pressure is 10^5 pascal. Mark the correct alternative(s).

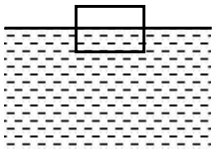


(A) v depends on h_1 and h_2 only but not on d

(B) If h_1 is decreased and h_2 and d both are gradually increased then v will also increase.

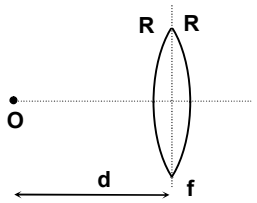
(C) Maximum speed with which the liquid can come out is 14.14 m/s.
(D) Maximum speed with which the liquid can come out is 10 m/s

10. A solid cubical body of density ρ_S is floating half submerged in a liquid of density ρ_L at temperature T. If γ_S and γ_L are coefficient of volume expansion of solid and liquid respectively and $\gamma_L > \gamma_S$. Identify the correct option(s).



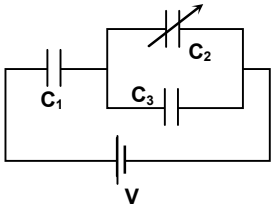
(A) $2\rho_S = \rho_L$, at temperature T
(B) If temperature of the system is decreased then the fraction of volume of solid submerged in liquid decreases.
(C) If depth of the block submerged in the liquid does not change on decreasing temperature then $2\gamma_S = 3\gamma_L$
(D) If the block does not expand on heating and the liquid contracts on heating then the block begins to sink in the liquid.

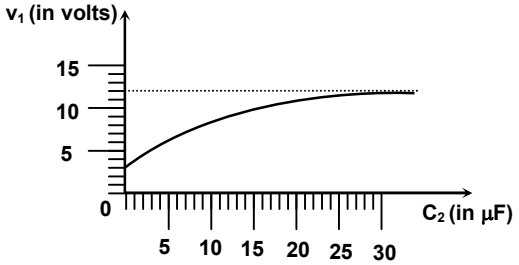
11. A thin equiconvex lens made of a material having refractive index μ_L = 3/2, has a focal length f in air. Image of an object placed in front of it, is inverted, real and magnified. Now the whole arrangement is immersed in water ($\mu_w = \frac{4}{3}$) without changing distance between the object and the lens. Then



(A) The focal length of lens relative to water will be 4f.
(B) New image will be virtual, erect and magnified.
(C) New image will be real, inverted and diminished.
(D) If this lens is cut laterally and then longitudinally into four identical parts (inside water) then the focal length of each part will be 8f.

12. In the given circuit C_2 is a variable capacitor. The graph represents the variation in potential difference (V_1) across C_1 when the capacitance C_2 is gradually increased from 0 to ∞. It is also observed that as $C_2 \to \infty$, V_1 approaches 12 V.



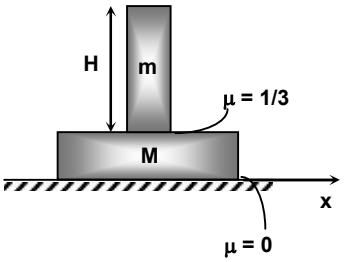


(A) Potential difference (V) across the battery is 12 Volt.
(B) $\frac{C_1}{C_3}$ has a value of 3.
(C) As C_2 is increased the energy stored in C_1 increases
(D) As C_2 is increased the energy stored in capacitor C_3 increases.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A solid cylinder of mass m, height H and diameter H/4 is placed on a plank of mass M as shown in the figure. The coefficient of friction between the plank and cylinder is 1/3 and that between the plank and ground is zero. The plank is made to oscillate along x-axis as given by the equation x = Asin(10t). Find the maximum amplitude of oscillation (in cm) for which the cylinder remains in equilibrium with respect to the plank.

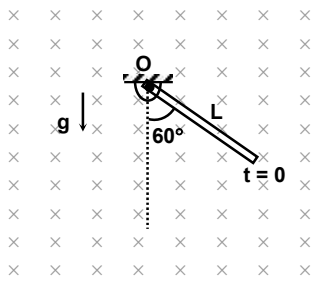


14. A soap bubble of radius r and surface tension T is formed in vacuum. It is slowly sprayed with some amount of charge, as a result of which it slowly expands and finally stops expanding when radius becomes 2r. If the amount of charge given to the bubble is $N\sqrt{12\pi^2 r^3 T\varepsilon_0}$, then find N. (assume temperature of the gas to be constant during expansion)

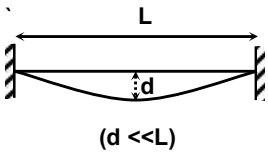
15. Peak emission from a black body at a certain temperature occurs at a wavelength of 4000 Å. On increasing its temperature, the total radiation emitted is increased to 256 times. These radiations are allowed to fall on a metal surface. The photoelectrons emitted by peak emission wavelength at higher temperature can be brought to rest by applying a potential which is equivalent to the excitation potential corresponding to the transition from level n = 2 to level n = 4 in the singly ionised Helium atom (obeys Bohr's Model). The work function of the metal (in eV) is

[Given E(in eV) = $\frac{12400}{\lambda(\text{in Å})}$]

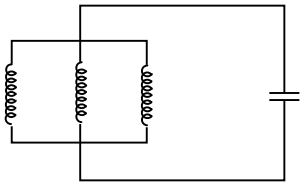
16. A non conducting uniform rod of mass 1 kg and length 1 m is hinged at one end as shown in the figure. A charge q = 1 C is uniformly distributed over its length. The rod is released from rest in a uniform magnetic field of $2\sqrt{15}$ T at time t = 0 from the position shown. Find the hinge reaction (in Newton) when the rod becomes vertical. (Take g = 10 m/s^2)



17. A very thin wire having mass per unit length μ is kept in tension between two fixed vertical walls at a separation (L = 10m). Due to its own weight the wire sags downwards in the middle by a small amount d = 8 mm. Assuming tension to be constant along the wire (due to its small mass). The speed of transverse wave in the wire can be expressed by N × 10^2 m/s. Find N. (Take g = 10 m/s^2)



18. A circuit consists of three coils of inductance 6mH each and 10 μF capacitance. At an instant when the charge on the capacitor is zero, currents 1.0 A, 2.0 A and 4.0 A are flowing in the same direction in these three coils. Find the maximum charge (in μC) stored in the capacitor. (Given $\sqrt{2} = 1.414$)



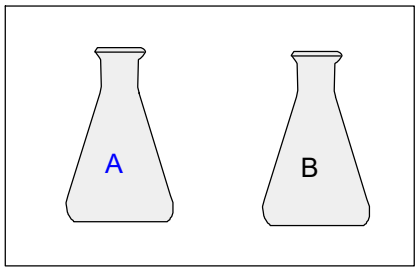
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

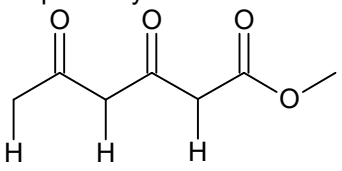
19. Aqueous solution of glucose and compound X are place in beaker A and B respectively in closed chamber.



'A' contains 40% (w/W) $C_6H_{12}O_6$ aq. Solution and 'B' contains 10% aq. (w/W) Solution of X (non-electrolyte). What is the molecular mass of compound X?

(A) 30
(B) 40
(C) 50
(D) 35

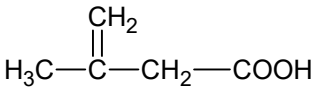
20. x, y and z are marked for hydrogen atoms in the following molecule which have K_a values respectively as

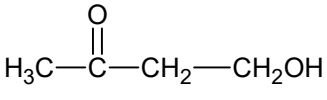


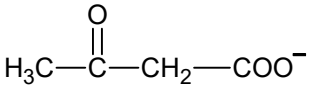
x y z

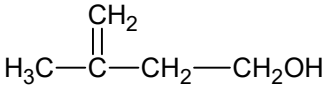
(A) 10^{-9}, 10^{-20} and 10^{-13}
(B) 10^{-20}, 10^{-9} and 10^{-13}
(C) 10^{-20}, 10^{-13} and 10^{-9}
(D) 10^{-9}, 10^{-13} and 10^{-20}

21. $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-COOH \xrightarrow{1\ \text{equiv. } Ph_3P=CH_2}$ Product will be:

(A) 

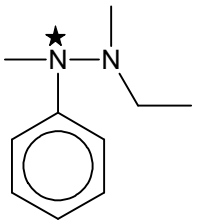
(B) 

(C) 

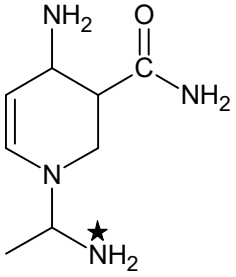
(D) 

22. In the following compound (*) marked atom is most basic in that compound. Identify the correct representation?

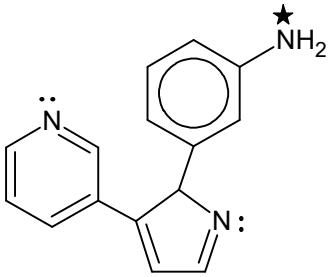
(A)



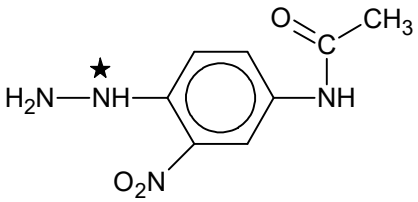
(B)



(C)



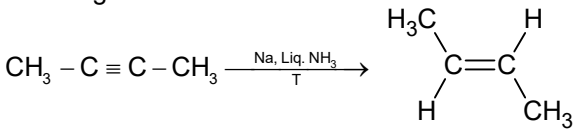
(D)



(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. For the given reaction:

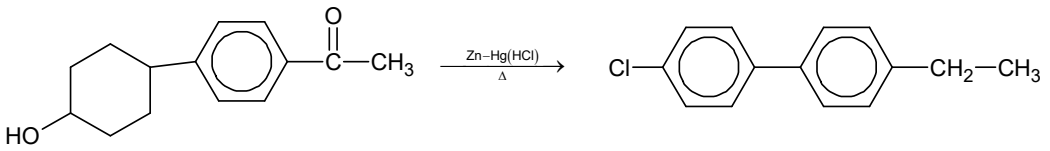


The correct statement(s) for the reaction mechanism and conditions is(are):

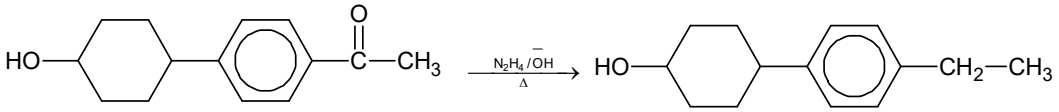
(A) Low temperature is required to keep NH_3 in liquid state by using dry ice/acetone.
(B) Radical anion is formed.
(C) Vinyl radical is formed.
(D) Vinyl anion is formed.

24. Which reactions are correct as per the given conditions?

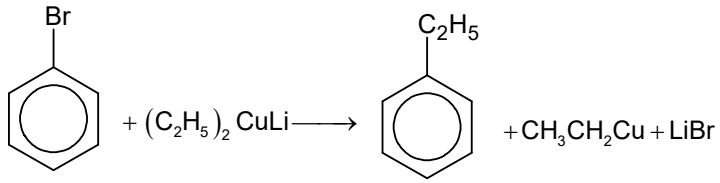
(A)



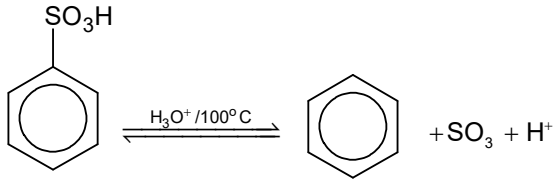
(B)



(C)



(D)



25. Identify the correct statements?

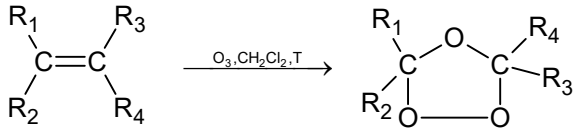
(A) Hydrolysis of sucrose brings about a change in sign of rotation, from laevo (–) to dextro (+) and the product is named as invert sugar.

(B) Maltose composed of two α-D glucose and α-D galactose units.

(C) Lactose is more commonly known as milk sugar.

(D) Amylopectin is insoluble in water.

26.



The above reaction involves

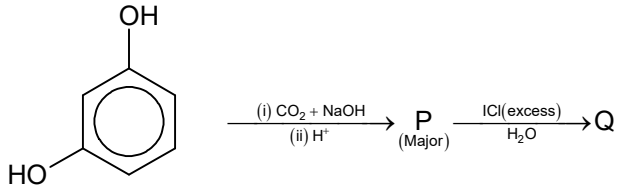
(A) Reaction occurs at 25°C.

(B) Reaction occurs below – 50°C.

(C) Formation of peroxy Zwitter ion take place.

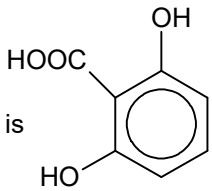
(D) It is an example of 1, 3-dipolar addition.

27.

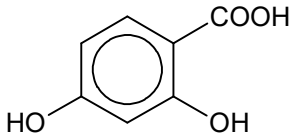


(A)

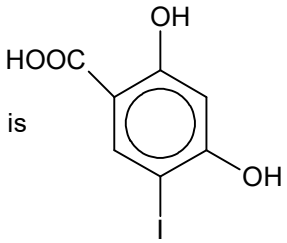
Compound P is



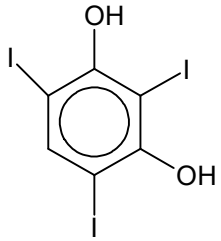
(B) Compound P is



(C) Compound Q is



(D) Compound Q is



28. Optical isomerism is/are possible in
(A) $[Cr(C_2O_4)_3]^{-2}$
(B) $[Pt(en)_3]^{+3}$
(C) $[Co(en)_2(NH_3)Cl]^{+2+}$
(D) $[Pt(Py)_2(NH_3)_2Cl_2]^{+2}$

29. Identify the correct reaction:
(A) $NiSO_4 + 2DMG + 2NH_4OH \longrightarrow \left[Ni(DMG)_2\right] + (NH_4)_2 SO_4 + 2H_2O$
(B) $2H_2S + SO_2 \longrightarrow 2H_2O + 3S$ (Colloidal sulphur)
(C) $4FeCl_3 + 3K_4\left[Fe(CN)_6\right] \longrightarrow Fe_4\left[Fe(CN)_6\right]_3 + 12KCl$
(D) $2CrO_2Cl_2 + 4NaOH \longrightarrow 2NaCl + Na_2Cr_2O_7 + 2H_2O$

30. Identify correct statements:
(A) Cl_2O is used as a bleaching agent for paper pulp and textile.
(B) Iodine oxides I_2O_4, I_2O_5 and I_2O_7 are insoluble solids in water.
(C) SO_2 is used in bleaching wool and silk.
(D) Pure ozone in violet black in solid state.

SECTION – C
(Numerical Answer Type)

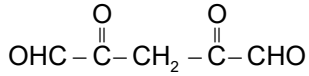
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. In the given compounds number of emulsion and gels are x and y respectively.
Butter, milk, jam, curd, shampoo, polish, fog, mist, rubber, smoke, blood, cheese, jellies.
Find the value of x/y?

32. $\text{Compound}(A) \xrightarrow[\text{(ii) } H_2O+Zn]{\text{(i) } O_3} \text{Products}$

(alkene)

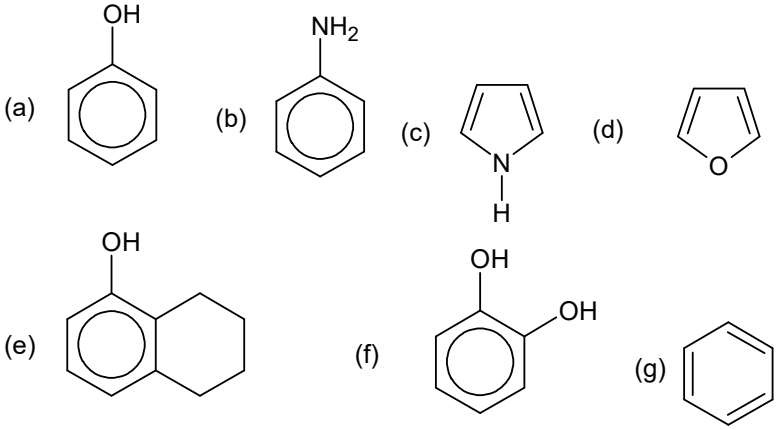
Ozonolysis of 1 mole compound A gives in products 2 mole CH_3CHO, 2 mole CHO – CHO, 1 mole OHC – CH_2 – CHO and 1 mole



If a= number of sigma bonds in compound A
b= number of pi bonds in compound A

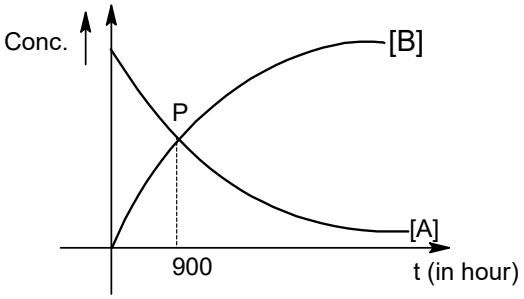
find the value of $\frac{a}{(b+2)}$?

33. In the following compounds, find the ratio of total number of compound which give Reimer Tiemann reaction with the total number of compound which do not give Reimer Tiemann reaction.



34. In the following compunds some undergo disproportionation with conc. NaOH and some do not give disproportionation with conc. NaOH.
F_2, Cl_2, Br_2, N_2, CH_3COCH_3, CCl_3CHO, $(C_6H_5)_3CCHO$, P_4, I_2, $(CH_3)_2CHCHO$, HCHO.
'x' are the number of compound which undergo disproportionation and 'y' are the number of compound which do not undergo disproportionation then x/y is

35. Concentration vs time graph is given below for first order reaction $7A \longrightarrow B$



Half-life for A in days is.

36. Find the value of $\frac{x+y}{2}$ if

(i) x is number of compound which contain nitrogen in the following
(Chile salt petre, Lunar caustic, Borax, Milon's base)
(ii) y is number of compound which evolve CO_2 on heating.
(Ag_2CO_3, Na_2CO_3, $(NH_4)_2CO_3$, $CaCO_3$, $NaHCO_3$, Hg_2CO_3.)

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Let $f(x)$ be a differential function such that $f^2(x)+xf(x)=3$, then $\int \frac{3x^3+6x^2f(x)+2f(x)}{(2f(x)+x)\left(x^3-2f(x)\right)^2}dx$ is equal to

(A) $\frac{1}{x^3-2f(x)}+c$

(B) $\frac{1}{2f(x)-x^3}+c$

(C) $\frac{1}{2f(x)+x}+c$

(D) $\frac{1}{x^3+2f(x)}+c$

38. A gambler has one rupee in his pocket. He tosses an unbiased normal coin unless either he is ruined or unless the coin has been tossed for a maximum of five times. If for each head he wins a rupee and for each tail he looses a rupee, then the probability that the gambler is ruined is

(A) $\frac{1}{2}$

(B) $\frac{5}{8}$

(C) $\frac{3}{8}$

(D) $\frac{22}{32}$

39. 5 out of 6 persons who usually work in an office prefer coffee in the mid morning, the other one always drink tea. This morning of the usual 6, only 3 are present. The probability that one of them drinks tea is

(A) $\frac{1}{2}$

(B) $\frac{1}{12}$

(C) $\frac{25}{72}$

(D) $\frac{5}{72}$

40. If the quadratic equation $(2m+1)x^2+(2n+1)x+(2p+1)=0$, where $m,n,p \in I$, has real roots then both roots are
(A) Rational
(B) Irrational
(C) Positive
(D) Can't say

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. If the value of $\left(\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1+2r}{r^4+2r^3+r^2+1}\right)\right)$ is α, then
(A) $\sin\alpha=\frac{1}{2}$
(B) $\sin\alpha=\frac{1}{\sqrt{2}}$
(C) $\cos\alpha=\frac{\sqrt{3}}{2}$
(D) $\cos\alpha=\frac{1}{\sqrt{2}}$

42. If $L=\lim\limits_{n\to\infty} n^{-n^2}\left((n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right).....\left(n+\frac{1}{2^{n-1}}\right)\right)^n$ and $|k|=\ln L$, then possible value(s) of k is/are
(A) –1
(B) –2
(C) 1
(D) 2

43. If ω is the imaginary cube root of unity such that $\left|\left(\sum\limits_{r=1}^{n}\left(r\sum\limits_{p=1}^{r}\left(\omega^{p-1}\right)\right)\right)-155\omega\right|=\left(\sum\limits_{r=1}^{n}\left(r\sum\limits_{p=1}^{r}\left(\omega^{p-1}\right)\right)\right)-155\omega$, then the value of n is equal to
(A) 29
(B) 30
(C) 31
(D) 32

44. If differential equation corresponding to family of curves $y=A\cos 2x+B\sin^2 x+C$ is given by $\lambda\frac{d^3y}{dx^3}+f(x)\frac{d^2y}{dx^2}=2\frac{dy}{dx}\cos 2x$ where λ is a real constant and $f(x)$ is some function in x, then
(A) $\lambda=0$
(B) $\lambda=1$
(C) Number of solutions of the equation $f(x)=\lambda$ in $(0,4)$ are 2
(D) Number of solutions of the equation $f(x)=\lambda$ in $(0,4)$ are 3

45. The extreme value (s) of the expression, $\frac{|\sec x|}{\sqrt{2}}\left[\sqrt{-2-4\cos 2x}+2\sqrt{2}\sqrt{2+3\cos 2x}\right]$ is/are

(A) $\sqrt{2}$
(B) $2\sqrt{2}$
(C) $\sqrt{10}$
(D) none of these

46. The function $\frac{\sin(x+a)}{\sin(x+b)}$ has no maxima or minima if

(A) $b - a = n\pi,\ n \in I$
(B) $b - a = (2n + 1)\pi,\ n \in I$
(C) $b - a = 2n\pi,\ n \in I$
(D) none of these

47. Let $A=\begin{bmatrix}1 & 2 & 1\\ 2 & 2 & 3\\ 1 & 1 & 2\end{bmatrix}$, then

(A) $A^3 - 5A^2 + I = 0$
(B) $A^{-2} = 5I - A$
(C) $-A^{-1} = \text{adj}A$
(D) A^3 is non-invertible matrix

48. If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of straight lines whose slopes are m and m^2, then the value(s) of 'a' can be

(A) -8
(B) 8
(C) 27
(D) -27

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If $f(k)=\lim\limits_{x\to 0}\sum\limits_{r=1}^{k^2}\frac{\ln(1+rx)}{x}$, $g(k)=\lim\limits_{x\to 0}\sum\limits_{r=1}^{k}\frac{e^{r^3x}-1}{x}$, then the value of $\lim\limits_{x\to\infty}\frac{f(x)}{g(x)}$ is

50. If $\alpha=\lim\limits_{k\to\infty}\left(\frac{k+2}{k^2+k+1}+\frac{k+4}{k^2+2k+4}+\frac{k+6}{k^2+3k+9}+.....+\frac{5}{7k}\right)$, then the value of e^{α} is

51. Let $y = f(x)$ be a differential function satisfying $\int\limits_2^x f(t)dt + 2 = \frac{x^2}{2} + \int\limits_x^2 t^2 f(t)dt$, then the value of $\int\limits_{-\pi/4}^{\pi/4}\frac{f(x)+x^9-x^3+x+1}{\cos^2 x}dx$ is equal to

52. Let f be a function such that $f:R\to R$, $f(x)=\dfrac{3x^2+mx+n}{x^2+1}$ and the range of $f(x)$ is $\left[-4,3\right)$, then the value of m^4+n^4 is equal to

53. If y satisfies the differential equation $xdy=y\left(xy^3+1\right)dx$ and $y(1)=2$, then the value of $\lim\limits_{x\to\infty} y$ is equal to

54. The area bounded by the circles $x^2+y^2=1$, $x^2+y^2=4$ and the pair of lines $\sqrt{3}\left(x^2+y^2\right)=4xy$ is equal to